OPERATING SYSTEMS GOALS, FUNCTIONS AND SERVICES

Goals and Functions of OS

OS can further be described by what they do i.e. by their functions, goals and objectives. Therefore, we will quickly run you through some of the goals of the OS which are:

Convenience for the User

When there is no OS, users of computer system will need to write machine-level program in order to manipulate the hardware. With OS, users can now easily and conveniently use the computer with no stress of directly programming the hardware. OS provide a convenient interface for using the computer system.

Efficiency

An OS allows computer system resources to be used in an efficient manner. This particularly important for large shared multi-user systems which are usually expensive. In the past, the efficiency (i.e. optimal use of the computer resources) considerations were often more important than convenience.

Evolutionary Capabilities

Ability to evolve also happens to be one of the goals of the OS. An OS should be constructed in such a way that it permits the effective development, testing and introduction of new system functions without interfering with its service.

Views of OS

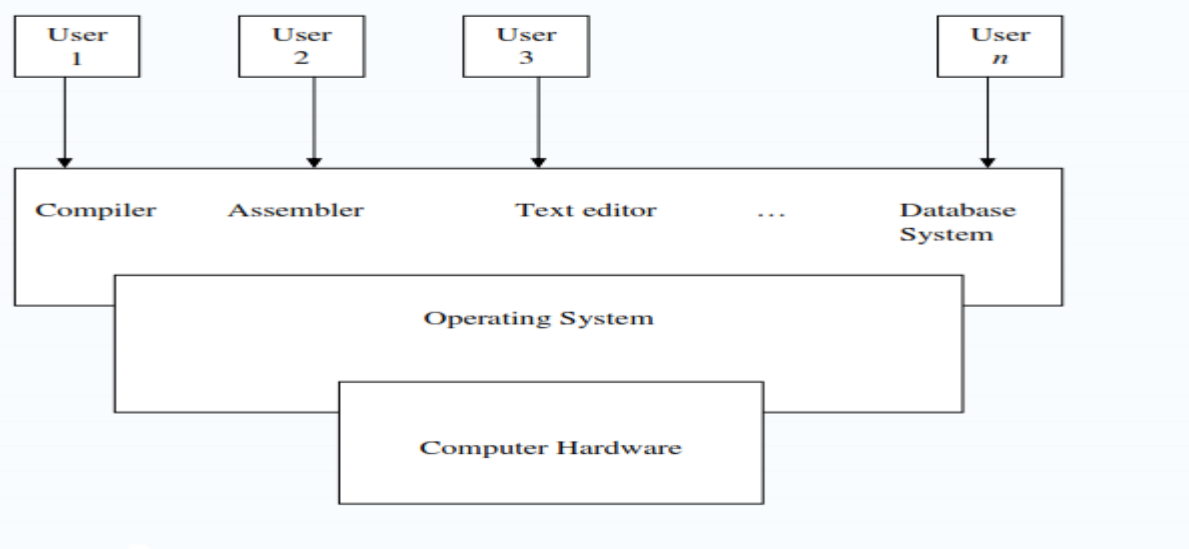
OS can be viewed from the perspective of what they are. These views are diverse depending on the particular view point of a user. Some of these views are discussed below.

OS as a User/Computer Interface

A computer system can be viewed as a layered or hierarchical structure consisting of the hardware, operating system, utilities, application programs and users. The users of application programs are called the end-users and are generally not concerned with the computer's architecture. The end-user views the computer system in terms of an application.

The application is developed by the application programmer who uses a programming language and a language translator. A set of programs called the utilities is provided to assist the programmer in program creation, file management and the control of Input/Output (I/O) devices.

The most important system program, operating system masks the details of the hardware from the programmer and provides a convenient interface for using the system. it acts as a mediator, making it easier for the programmer and for application programs to access and use the available services and facilities.



Abstract View of the Components of a Computer System

The following are some of the important functions of an operating System.

1. Memory Management
2. Processor Management
3. Device Management
4. File Management
5. Security
6. Control over system performance
7. Job accounting
8. Error detecting aids
9. Coordination between other software and users

Memory Management

Memory management refers to management of Primary Memory or Main Memory. Main memory is a large array of words or bytes where each word or byte has its own address. Main memory provides a fast storage that can be access directly by the CPU. So for a program to be executed, it must in the main memory. Operating System does the following activities for memory management.

- Keeps tracks of primary memory i.e. what part of it are in use by whom, what part are not in use.
- In multiprogramming, OS decides which process will get memory when and how

much.
- Allocates the memory when the process requests it to do so.
- De-allocates the memory when the process no longer needs it or has been terminated.

Processor Management

In multiprogramming environment, OS decides which process gets the processor when and how much time. This function is called process scheduling. Operating System does the following activities for processor management.

- Keeps tracks of processor and status of process. Program responsible for this task is known as traffic controller.
- Allocates the processor (CPU) to a process.
- De-allocates processor when processor is no longer required.

Device Management

OS manages device communication via their respective drivers. Operating System does the following activities for device management.

- Keeps tracks of all devices. Program responsible for this task is known as the I/O controller.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

File Management

A file system is normally organized into directories for easy navigation and usage. These directories may contain files and other directions. Operating System does the following activities for file management.

- Keeps track of information, location, uses, status etc. The collective facilities are often known as file system.
- Decides who gets the resources.
- Allocates the resources.
- De-allocates the resources.

Other Important Activities

1. **Security**-By means of password and similar other techniques, preventing

unauthorized access to programs and data.

2. **Control over system performance**-Recording delays between request for a service and response from the system.
3. **Job accounting**-Keeping track of time and resources used by various jobs and users.
4. **Error detecting**-Production of dumps, traces, error messages and other debugging and error detecting aids.
5. **Coordination between other software and users**-Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems.

A computer system has a set of resources for the movement, storage and processing of data. The OS is responsible for managing these resources. Note that resources include CPU, file storage space, data, programs, memory space, I/O devices, etc

The OS is like any other computer program in that it provides instructions for the processor. The key difference is in the purpose of the program. The OS directs the processor in the use of the other system resources and in the timing of its execution of other programs. The processor, in order to do any of these things, must cease execution of the OS program to execute other programs. Thus, the OS relinquishes control long enough to prepare the processor to do the next piece of work.

The portion of the OS that is always in main memory is called the kernel or nucleus and it contains the most frequently used functions in the OS. The remainder of the main memory contains other user programs and data. The allocation of this resource (i.e. main memory) is controlled jointly by the OS and the memory management hardware in the processor.

Services Provided by the OS

The services provided by the OS can be categorized into two:

Convenience for the Programmer/User

The conveniences offered the user are in diverse and following ways:

a) Program Creation: Although editors and debuggers are not part of the OS, they are accessed through the OS to assist programmers in creating programs.
b) Program Execution: OS ensures that programs are loaded into the main memory.
c) I/O devices and files are initialized and other resources are prepared. The program must be able to end its execution either normally or abnormally. In case of abnormal end to a program, it must indicate error.
d) Access to I/O devices: Each I/O device requires its own set of instructions or control signal for operation. The OS takes care of the details so that the programmer can think in terms of reads and writes.

e) Controlled Access: In the case of files, control includes an understanding of the nature of the I/O device (e.g. diskette drive, CDROM drive, etc.) as well as the file format of the storage medium. The OS deals with these details. In the case of the multi-user system, the OS must provide protection mechanisms to control access to the files.
f) Communications: There are many instances in which a process needs to exchange information with another process. There are two major ways in which communication can occur:
   - It can take place between processes executing on the same computer.
   - It can take place between processes executing on different computer systems that are linked by a computer network.

   Communications may be implemented via a shared memory or by a technique of message passing in which packets of information are moved between processes by the OS.

g) Error Detection: A variety of errors can occur while a computer system is running.

   These errors include:

   o CPU and memory hardware error: This encompasses memory error, power failure, a device failure such as connection failure on a network, lack of paper in printer.

   o Software errors: Arithmetic overflow, attempt to access forbidden memory locations, inability of the OS to grant the request of an application.

   In each case, the OS must make a response that makes the less impact on running applications. The response may range from ending the program that caused the error, retrying the operation or simply reporting the error to the application.

Efficiency of System: Single and Multi-User

In the area of system efficiency, the OS offer the following services:

a) System Access or Protection: In the case of a shared or public system, the OS controls access to the system and to specific system resources by ensuring that each user authenticates him/herself to the system, usually by means of passwords to be allowed access to system resources. It extends to defending external I/O devices including modems, network adapters from invalid access attempts and to recording all such connections for detection of break-ins.
b) Resources Allocation: In an environment where there multiple users or multiple jobs running at the same time, resources must be allocated to each of them. Many different types of resources are managed by the OS. Some (such as CPU cycles, main memory and file storage) may have general request and release codes. For instances, in determining how best to use the CPU, the OS have CPU-scheduling

routines that take into account the speed of the CPU, the jobs that must be executed, the number of registers available and other factors. These routines may also be used to allocate plotters, modems and other peripheral devices.

c) Accounting: This helps to keep track of how much of and what types of computer resources are used by each user. Today, this record keeping is not for billing purposes but for simply accumulating usage statistics. These statistics may be available tool for researchers who want to reconfigure the system to improve computing services.
d) Ease of Evolution of OS: A major OS will evolve over time for a number of reasons such as hardware upgrades and new types of hardware e.g. The use of graphics terminals may affect OS design. This is because such a terminal may allow the user to view several applications at the same time through 'windows' on the screen. This requires more sophisticated support in the OS.
e) New Services: In response to user demands or the need of system managers, the OS may expand to offer new services.
f) Error correction: The OS may have faults which may be discovered over the course of time and fixes will need to be made.

Other features provided by the OS includes

- Defining the user interface
- Sharing hardware among users
- Allowing users to share data
- Scheduling resources among users
- Facilitating I/O
- Recovering from errors